Wang 2019 GBM — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

9,916 malignant → 9,650 (7/10 samples, ≥237) → 9,634 on one connected component.
Sample iLISI rose 1.0→2.6/7 under Harmony (plateau by θ=10); one 16-cell island removed.

Gate 2 — module geography → GO (non-branching)

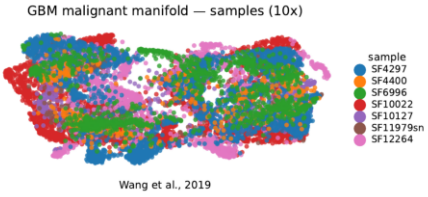
Shared Gast/pSom gradient (r=+0.16, co-vary) with Crest an inverse terminal readout (r=−0.49 vs Gast, −0.87 vs pSom). Curve fold-free (0 forks); Crest-high decile at pseudotime 0.85.

Gate 3 — hub cross-repression → NA

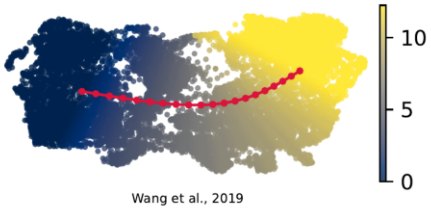
Gast/pSom co-vary (not compete) → single collapsed axis, no bifurcation to test.

STEP FIGURES

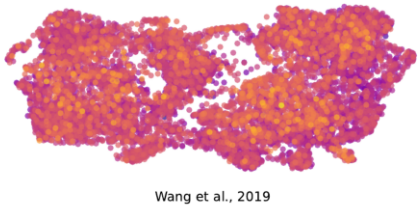
Gate 1: integration



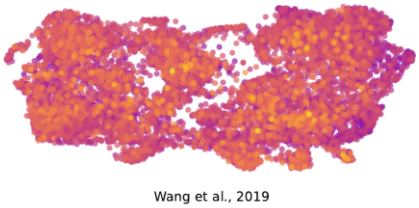
Trajectory / root



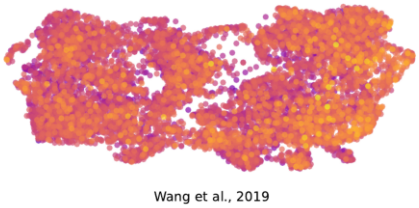
Gast



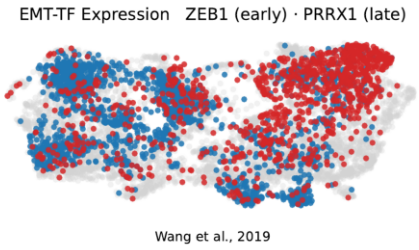
pSom



Crest



EMT-TFs



Dataset. Glioblastoma (Wang et al., 2019); 10x, 10 samples; 9,916 malignant raw → 9,634 final after threshold + island removal. IDH-wildtype primary GBM (single 10x leaf).

Verdict. ✓ Pattern reproduces as a non-branching embryonic axis: pSom-high origin → Crest-high terminus (pSom ρ=−0.62, Crest ρ=+0.49 along pseudotime; Gast flat, ρ=−0.06), continuity confirmed on a diffusion embedding independent of the fitted UMAP (r=−0.98). Linear (single collapsed axis; GBM states NPC/OPC→AC/MES span it). Not branching.

Parameter choices

Min cells/sample	≥237 (mult-gap 396→143; 7/10)	**Integration**	Harmony θ=10; PCA20; spread=1.3
Geometry	Linear / non-branching (curve)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	pSom-high / Crest-low tip	**EMT-TFs**	ZEB1 (early/root) → PRRX1 (late)